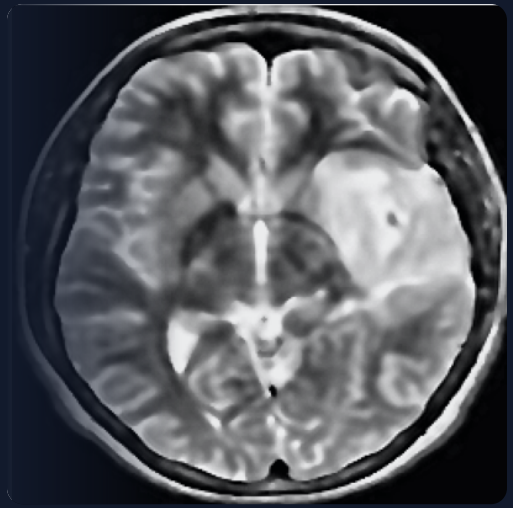
DIGITAL SIGNAL PROCESSING · GROUP 11

Medical Image Enhancement and Diagnostic Assistance for Brain Tumor Detection

Enhancing brain MRI with OpenCV, marking the suspicious region, and classifying tumor vs no tumor, and which kind, with every stage shown so a clinician can check it



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THE PROBLEM

The tumor is in the MRI. Noise and low contrast hide it, and a bare label can’t be checked.

Barrier 1

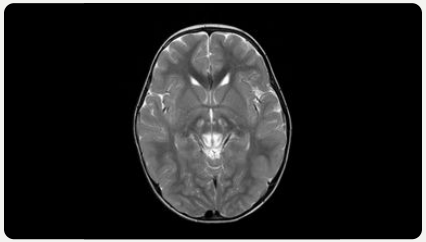
The image hides the lesion

MRI slices carry noise, low contrast and unclear boundaries. A low-contrast tumor next to bright skull is hard to see, and reading many scans tires even trained radiologists.

Barrier 2

A label alone can't be verified

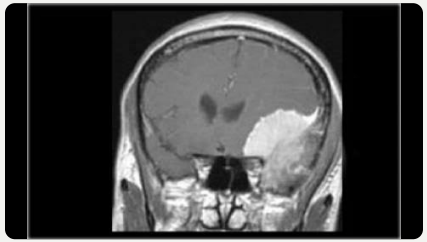
A classifier that only says "tumor, 97%" gives the radiologist nothing to inspect. A useful assistant has to show where it looked and what it measured.



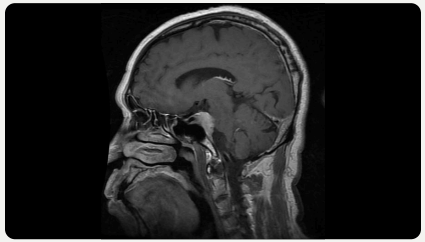
No tumor · 500 slices



Glioma · 926 · grows inside the brain tissue



Meningioma · 937 · arises from the membranes at the rim



Pituitary · 901 · on the midline, at the skull base

Our aim: enhance the slice, locate the suspicious region, classify it, show every stage, and measure each claim on held-out data.

THE SIGNAL

An MRI slice is a 2-D signal, and its noise is not Gaussian

A magnitude MRI image is |complex signal + complex noise|, so the noise is Rician: biased upwards in dark regions. Scanned and exported slices also carry impulse (salt-and-pepper) pixels. Each needs a different filter.

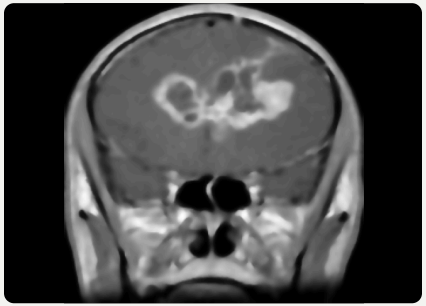
$$\text{Rician magnitude:}\quad m = \sqrt{(S + n_1)^2 + n_2^2},\quad n \sim N(0, \sigma)$$

$$\text{Bilateral:}\quad BF[I]_p = (1/W_p)\ \Sigma_q\ G_{\sigma s}(\|p-q\|)\ \cdot\ G_{\sigma r}(|I_p-I_q|)\ \cdot\ I_q$$

The bilateral weight falls with distance **and** with intensity difference, so it smooths inside a tissue but stops at an edge.



As acquired



After median + bilateral: grain gone, lesion edges kept

Step 1 · nonlinear

Median 3×3

An order statistic: an impulse never reaches the middle of the sorted window, so it vanishes without blurring edges.

Step 2 · edge-preserving

Bilateral

d = 7, σ_{colour} = 50, σ_{space} = 50. Smooths the remaining grain while keeping the tumor boundary sharp.

Before that

Same input every time

Grayscale, crop blank borders, resize to 256×256 (area interpolation when shrinking).

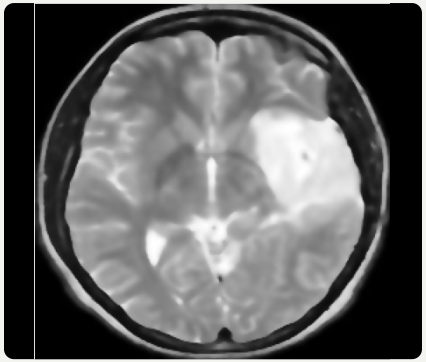
After that

Min-max normalise

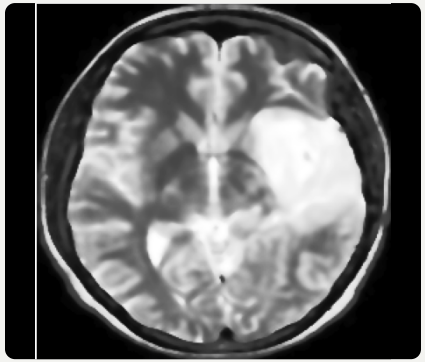
Stretch to 0–255 so every scanner's slice starts from the same range.

ENHANCEMENT

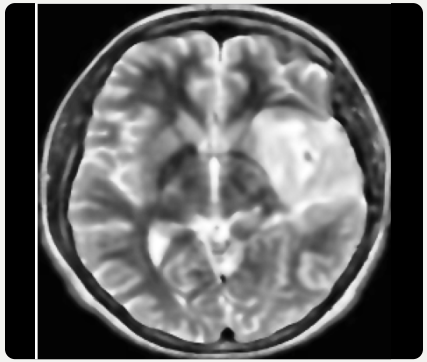
CLAHE lifts local contrast without flooding the image



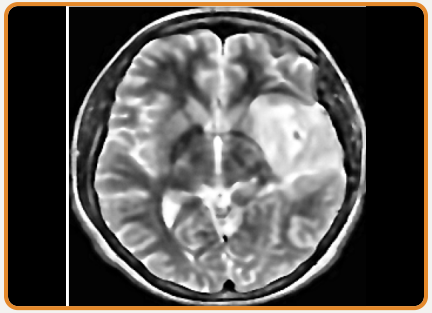
Preprocessed input



Global histogram eq. One mapping for the whole slice: bright skull and background dominate it.



CLAHE 8×8 tiles (32 px), clip limit 2.0: each tile equalised, noise held back by the clip.



CLAHE + unsharp mask (σ = 1.5, amount 1.2): the shipped output.

Mean over the 394 test images	Preprocessed	Global HE	CLAHE only	CLAHE + unsharp
Contrast (std of intensities)	57.9	**75.6**	65.3	70.4
Entropy (bits)	6.77	6.55 ↓	**7.41**	7.39
Sharpness (Laplacian variance)	338	808	391	**1428**

Global HE gives the most raw contrast but loses information (entropy falls below the input). CLAHE + unsharp raises all three.

THE SYSTEM

Seven stages, from an uploaded slice to a checked verdict

S1
Data
Public 4-class MRI set, 3,264 slices, its own train / test split. A synthetic MRI phantom for tests.

S2
Preprocess
Grayscale, 256×256, median + bilateral, normalise.

S3
Enhance
CLAHE + unsharp mask, with contrast, entropy and sharpness measured.

S4
Segment
Skull strip, statistical threshold, left-right asymmetry, morphology, contours.

S5
Features
171 numbers: intensity, texture, shape, asymmetry, location.

S6
Classify
SVM-RBF for detection. SVM + ResNet-18 blend for typing.

S7
Present
8-panel figure, REST API, web app with every stage and measurement.

Built in **Python**: OpenCV, scikit-image, scikit-learn, PyTorch, FastAPI. Web app: **Next.js 16**, TypeScript, Tailwind. Data: **Bhuvaji et al.** brain tumor MRI (No Tumor, Glioma, Meningioma, Pituitary): 2,870 train / 394 test.

Detection stays fully classical, so its evidence is inspectable. The CNN is added only where the classical features measurably fall short (typing). 66 automated tests cover the pipeline, API and CNN branch.

SEGMENTATION

Five steps from a slice to one suspicious region

1
Skull strip
Otsu head mask, largest component, holes filled, then peel 22% of the radius by distance transform.

→

2
Threshold
Tissue above the brain median: flag pixels ≥ mean + 1.5σ.

→

3
Asymmetry
Mirror about the brain's centroid; excess ≥ mean + 2σ joins the mask.

→

4
Morphology
5×5 ellipse: open ×2 removes specks, close ×3 fills gaps.

→

5
Contours
Drop <120 px or aspect >5; keep the largest × brightest.

Why not Otsu inside the brain? Otsu always splits a histogram in two, so on a healthy brain it returned grey vs white matter as a "lesion". On 60 scans it flagged **100%** of healthy brains; the tissue-relative rule flagged **3%** and still caught every tumor.

Why asymmetry? A tumor breaks the brain's left-right symmetry even when it is not the brightest tissue. With both rules the pipeline localised **129 of 155** tumors on the 253-image classic dataset.

FEATURES

Each scan becomes 171 numbers, in seven families

33 features

Intensity + histogram

Mean, std, skew, kurtosis, entropy, percentiles over brain pixels; a 16-bin histogram.

38 features

GLCM texture

Contrast, homogeneity, energy, correlation… at distances 1–3 px, 4 angles, 32 grey levels.

28 features

LBP texture

Uniform local binary patterns, radius 3, 24 points: 26 bins + entropy + uniformity.

12 features

Shape of the region

Area, perimeter, circularity $4\pi A/P^2$, solidity, extent, brightness vs brain.

10 features

Asymmetry

Statistics of the mirrored difference and its largest blob.

20 features

Regional

Mean and std in a 3×3 grid over the brain.

30 features

Location + plane

Distance from the midline, depth, rim contact, side; scan-plane shape cues. Scale-free.

Why location

Tumor type follows place

Pituitary sits on the midline, meningioma at the rim, glioma inside. Texture-only typing reached 72.1%.

THE CLASSIFIERS

Four classical models compete; a CNN joins only for typing

Every model sees the same 171 features (StandardScaler inside the pipeline, so cross-validation never leaks scale). The winner is chosen by 5-fold stratified cross-validated ROC-AUC on the training split only.

0.984 / 0.962

Logistic regression

C = 1, balanced class weights. The linear baseline.

0.993 / 0.983

SVM-RBF · selected

C = 10, γ = scale, calibrated probabilities. Won both tasks.

0.987 / 0.967

Random forest

400 trees, min 2 samples per leaf, balanced.

0.990 / 0.979

MLP

Hidden layers 128 → 64, ReLU, $\alpha = 10^{-3}$.

typing only

ResNet-18

ImageNet weights, all layers fine-tuned 14 epochs at 224×224, augmentation, 4-view test-time averaging.

CV ROC-AUC shown as detection / typing. **Detection threshold** = the lowest that keeps out-of-fold specificity ≥ 80%: `0.153`. **Blend weight** for typing chosen on a 431-image validation split: 0.95 on the CNN.

RESULTS · DETECTION

Detection: 271 of 289 tumors found, no healthy scan flagged

95.4%
accuracy on the 394 held-out test slices

93.8%
sensitivity: 18 tumors missed out of 289

100%
specificity: 105 of 105 healthy scans cleared

0.997
ROC-AUC (threshold-free)

SVM-RBF, 171 features	default cut	tuned 0.153
Missed tumors	33	**18**
False alarms	0	0
Accuracy	91.6%	**95.4%**
Balanced accuracy	94.3%	**96.9%**

In screening a missed tumor costs far more than a second look, so the threshold is tuned for **sensitivity**, and it was picked from out-of-fold training predictions, never from the test set.

Caution: all 105 healthy test slices have a near-twin in training, and the 18 missed tumors are all slices without one (next two slides).

THE RESEARCH GAP, MEASURED

99% → 44%

Same typing model, same test folder. 259 of the 394 test slices have a near-identical twin in the training folder (88 are byte-for-byte copies). On those it scores 99.2%; on the 135 slices with no twin, 43.7%. Most of the benchmark score comes from recognising images it has already seen.

Our duplicate audit: 64×64 normalised cross-correlation ρ ≥ 0.98, as in Vangala et al. · 10

RESULTS · DUPLICATE AUDIT

Split the test set by “seen before”, and the weak spots appear

Held-out test slices	twin in training 259 slices	no twin 135 slices
Detection sensitivity	100% (154/154)	**86.7%** (117/135)
Detection specificity	100% (105/105)	not measurable: 0 healthy
Typing accuracy	99.2%	**43.7%**
Glioma recall	90.0% (18/20)	**21.3%** (17/80)
Pituitary recall	100% (25/25)	75.5% (37/49)
Meningioma recall	100% (109/109)	83.3% (5/6)

Every one of the 18 missed tumors is a slice with no twin in training.

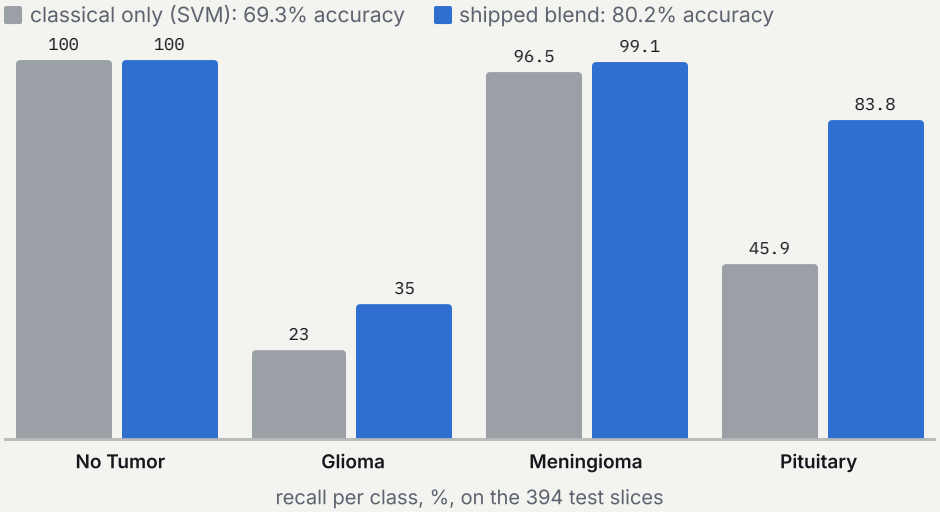
Every healthy test slice has a twin (88 are exact copies), so the 100% specificity says nothing about a healthy brain the model has not seen.

Glioma is weak because it is new: 80 of the 100 test gliomas have no twin, and every training glioma is 512×512 while the test gliomas come in 64 sizes.

Vangala et al. (arXiv:2610.00421, 2026) audited this same dataset and also found 259 near-twins; our count matches theirs exactly.

RESULTS · TYPING

Typing: the CNN carries it, and glioma stays hard



recall per class, %, on the 394 test slices

actual ↓ / predicted →	None	Gli	Men	Pit
No Tumor	**105**	0	0	0
Glioma	24	**35**	40	1
Meningioma	0	1	**114**	0
Pituitary	6	0	6	**62**

Of 100 test gliomas, 40 were called meningioma and **24 were called "no tumor"**. Precision on the glioma calls that were made is 97%. Macro F1 78.1%, one-vs-rest AUC 0.961. The validation split, cut from the training folder, reached 97.4% balanced accuracy; the test folder 79.5%.

THE OUTPUT

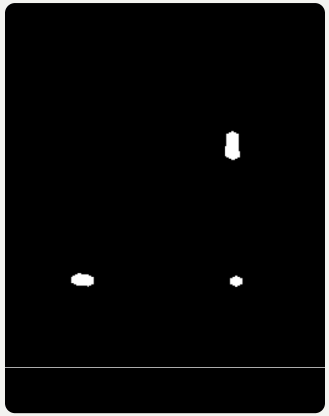
Every stage is shown, so the verdict can be checked



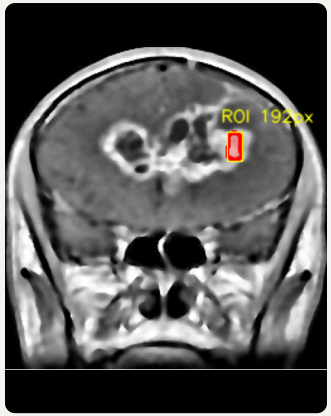
1 · Original as uploaded



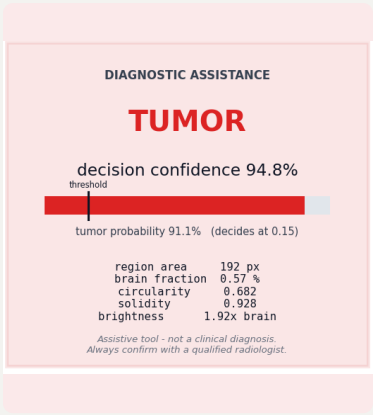
3 · Enhanced CLAHE + unsharp



4b · Mask threshold + morphology



5 · Region contour + box



Verdict probability 91.1%, threshold 0.15, region measurements

What the radiologist gets: the verdict, the probability against the threshold, the region's area, circularity, solidity and brightness, its location, and the enhancement gain. The web app shows the same, and uploads are never written to disk.

Read it critically: here the outlined region (192 px) covers only one bright part of the lesion. The verdict comes from all 171 features, not from the box, and that is exactly why the box is shown: a clinician can see when it is off.

ENGINEERING

Four real bugs, and what each one taught us

Bug 1 · segmentation

The flood fill saw half the frame as a "hole"

When the head touches the top and bottom borders, the background splits in two. A corner flood fill reached one half and the other half was filled in as brain. Fix: pad a 1-pixel black frame first.

Bug 2 · segmentation

Otsu flagged every healthy brain

Two classes are always found, so grey vs white matter became a "lesion": 100% of healthy scans flagged. Fix: a tissue-relative mean + 1.5σ rule (3% flagged).

Bug 3 · thresholding

Threshold tuning made results worse

Maximising specificity under a sensitivity floor picked 0.93 from over-confident training folds; test sensitivity fell from 91% to 71%. Fix: maximise sensitivity under a specificity floor.

Bug 4 · evaluation

The ensemble peeked at the test set

The first version kept whichever branch scored best on test, so the score was the best of three draws. Fix: choose the blend weight on a validation split, with 0 and 1 in the grid.

LIMITS

What these numbers do not show

Glioma typing is weak

35% recall. Detection is strong; typing is only partly solved.

A leaky public split

259 of 394 test slices have a near-twin in training. No patient IDs, so splits are by slice.

Specificity is untested

All 105 healthy test slices are near-copies of training slices; unseen healthy brains were never scored.

The CNN is a black box

It supplies most of the typing accuracy; the audit trail comes from the classical branch only.

2-D slices only

No 3-D volume, and intensity rules can be fooled by calcification or artefacts.

Not a medical device

An assistive triage tool; clinical use needs multi-centre validation and approval.

NEXT STEPS

What we would do next

Step	Why
De-duplicate and re-split	Hash and near-duplicate check before splitting, and a second dataset (e.g. Figshare, Cheng et al.) for an honest test
Grad-CAM over the CNN branch	Show which pixels drove the typing decision, so the CNN is checkable too
U-Net segmentation with Dice scoring	Replace the intensity rules with a pixel-level mask and measure overlap
Patient-level splits	Remove any chance that slices of one patient sit on both sides
3-D tumor volume	Combine slices of a full MRI stack into one volume estimate
Automated PDF reports	Hand the radiologist the stages, measurements and verdict as one document

THREE THINGS TO REMEMBER

1 Enhance locally: CLAHE + unsharp raised entropy 6.77 → 7.39 bits and sharpness 4×.

2 Pick thresholds away from the test set: missed tumors fell 33 → 18 with no false alarms.

3 Check the test set for twins: on the 135 slices never seen before, typing falls from 99% to 44%.

Thank you. Questions?